Air Circulation Robot

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Our motivation for the air circulation robot is to help create an environment that is healthier for people. Since the pandemic, the need for good air circulation has been increasing. People are more aware of the air quality around them, and the impact that bad air quality can have. With this robot, we can help increase air circulation and lower the risk of diseases like Covid-19 from spreading, mitigate allergic reactions, and assist with any other risks to respiratory health from particles in the air. This is an autonomous robot that can monitor and cleanse the air, acting like a Roomba for the air.

Changes & Deliverables

Since our initial project proposal, we've changed our design plan for the air circulation robot. Instead of using three wheels for our car, we will use four. We're also adding a relay module to increase the voltage, since some of the other components we are using require a higher voltage than what the Raspberry Pi 5 provides out of the box. We're also including an accelerometer sensor, which will be useful for helping us determine the relative location of our car over time. We plan on doing calculations based on the speed and time, which will let us know how far we've moved in one direction. Lastly, we're considering adding an LCD screen that will show our metrics as the car moves from one location to another. This way, our air circulation robot can act as a real-time monitor system as well as a circulation robot.

A major pivot is that we no longer intend to use an air filter as we concluded from some of our research that a better alternative would be to implement an air ioniser. Not only would this be more portable and lightweight, but it also would not frequently require filter replacements to function properly. We also found that it would be an interesting challenge for us to wire and interface with a negative ion generator module in order to implement this as it would involve more interaction with our software and sensors.

At the end we hope to have delivered a mostly autonomous robot that can map out the layout of the room (generally); monitor air quality through measurements of temperature, humidity, and air quality; and lastly to respond to data processing from sensors and software to improve the nearby ambient air conditions with a negative ion generator and fan.

Related Work

A big part of our project is being able to map out a room with an autonomous robot. There are plenty of people who have tried to do something similar. For example, Kai Nakamura made an autonomous mapping robot using SLAM. We are considering doing this, but with a lidar sensor. If we're able to get a decent mapping even with different sensors, that will be a big win for us.

In addition to autonomous navigation, we have found that negative ion generators have been used to improve air quality in various demonstrations such as this video showing that it can be effective at reducing the concentration of smoke in the air. There are also many examples online of people building automatic air monitoring systems or air conditioning systems using similar sensors and actuators that we are using (for example, in this video they apply similar sensors and actuators with a similar goal in mind).

However, we have not found many projects in which both autonomous navigation and automatic air monitoring/condition systems are integrated together, and this will likely be a major challenge as we put both components of the project together.

Development Progress

Status on the HW Component

So far in our development progress, we have been able to setup and test the ultrasonic sensor module, the digital temperature and humidity sensor module, and the PM2.5 air quality sensor module. We have adjusted the sensors and actuators that we intend to use so our current list of components is now:

- 1 Ultrasonic Sensor (37 in 1)
- 1 GY-521 Module (37 in 1)
- 1 USB to Serial Converter (Amazon)
- 1 PM2.5 Sensor and 2.54 Adapter Kit (Checkout part from CSE 147 Lab Inventory)
- 1 Temperature & Humidity Sensor (37 in 1)
- 4 DC Motors (Robot Chassis Kit from Amazon)
- 1 Motor Driver (Amazon)
- 4 Wheels (Robot Chassis Kit from Amazon)
- 1 Chassis (Robot Chassis Kit from Amazon / 3D-print our own)
- 1 Negative Ion Generator (Air Ioniser) Module (Amazon)
- 1 Fan (Amazon)
- 1 LCD Screen Display (37 in 1)
- 1 Button (37 in 1)
- 1 Rotary Encoder (37 in 1)
- 1 Portable Battery Pack (Amazon)

We have currently fully setup, implemented, and tested the main sensors we intend to use for our project, additional sensors and actuators were added with usability in mind during the development and testing process. We had initially hoped to have assembled more of the hardware components but found that a significant challenge was with configuring the sensors as many resources and libraries online require dependencies specific to Arduino. We found that porting the sensor capabilities to Raspberry Pi 5 to be more challenging than anticipated.

Status on the SW Component

Our goal for the mid-project report was to have the hardware sensors interacting perfectly with each other. However, we've also begun looking into the software component of our project. One big component is trying to map out a room with the robot, since we want to store information about the air quality at each location over time. Initially, we thought this would be easy. The idea was to map out one location of the room as the origin, and then increase the x and y coordinates of our location as our car moves along. After doing some research, this is harder than we thought. With our sensors, it will be hard to know exactly how far the car has traveled in one direction, as well as which direction the car is oriented towards. Mapping out the room will become even more difficult when we factor in obstacles, since there will be large parts of the room untouched. Right now, we're planning on creating a proof of concept with a car that moves around a room with no obstacles. We're planning on using a room with a predefined size, and simply storing these values in a 2d array. We'll use the accelerometer to help calculate how far we've moved in a direction, and the ultrasonic sensor to help us calibrate based on our distance to the wall. We've also looked into using SLAM (simultaneous localization and mapping) algorithms to help us map out a room. This would be a lot harder to do, but would allow us to map out a room the car has never seen before. Besides difficulty, the other big problem with this is our sensors might be inadequate to fully utilize SLAM.

Experimental Section

Our metrics of success are the same as before. We will measure success by checking the heat and air particle distribution before and after the device sweeps the room. We want the median and average to be close to one another. We also want the maximum and minimum temperature to be within 1 degree Fahrenheit of each other, and ideally reduce the range between the maximum and minimum concentration of air particles to be by at least 10%. If the air quality is observably more pleasant and desirable, then that would be the clearest indication of the success of the capabilities of our device.

We've tried using all the sensors to measure data. We found that some of the sensors are not too accurate, but will suffice for our purposes, which is a simple proof of concept. We have not tried experimenting to see if our robot will actually be able to purify a room but have successfully configured our Air Quality sensor and confirmed that it should be capable in sufficiently measuring air quality to a degree we can sufficiently demonstrate as proof of concept.

The biggest issue we have found with measuring data with the sensors is less so the accuracy but rather their responsiveness. While they can successfully report mostly accurate data, there are instances where they stall in their measurements in our program. We hope to experiment more with this as we put everything together and schedule all of the components of the program.

Final Demo

In the final demo, we will show what a room's temperature, humidity and air particles are like before and after. We will show the device moving around the room, and the data it collects as it goes along the room. We will report if we were able to meet or exceed our metrics. We will want to create certain environments to demonstrate our robot responding to certain air conditions (high temperature, high humidity, and poor air quality). If we fail to meet our metrics or our robot does not perform as desired, we will explain what went wrong, how we could improve it, and future steps to take to build upon our project in future implementations. We also want to report on our process, where we think we could have improved, and where we think we can take this project further moving forward.

Regarding what remains to be done, right now, we still need to choose how to implement the room mapping from one of the two ways we brainstormed. After that, we will need to try and see if we're able to improve the air circulation in our room with our robot. To do this we plan to have our robot assembled, all of the components (sensors and actuators) interfacing with one another successfully into one portable device, and ensure that they are able to interact with the software (notably with data processing such that we can arrange it in a manner that can be mapped and stored for future reference).